

Math Thermo

class # 08

02/12/2026

Free Energy Density

Suppose that $V > 0$ is fixed. Define, as before,

$$\rho := \frac{N}{V}, \quad \text{mole number density}$$

Then, since, for any $\lambda > 0$,

$$(8.1) \quad \lambda F(T, V, N) = F(T, \lambda V, \lambda N),$$

it follows that

$$\begin{aligned} \frac{1}{V} F(T, V, N) &= F\left(T, \frac{V}{V}, \frac{N}{V}\right) \\ &= F(T, 1, \rho). \end{aligned}$$

Now, define the free energy density

$$f = f(T, \rho) := F(T, 1, \rho).$$

It then follows that

(8.2)

$$F(T, V, N) = V f(T, \rho).$$

Recall, that Euler's Equation in density form is

(8.3)

$$\tilde{u}(s, \rho) = T_0(s, \rho) s - P_0(s, \rho) + \mu_0(s, \rho) \rho.$$

Thus,

$$F(T, V, N) = \tilde{U}(S_T, V, N) - TS_T$$
$$Vf(T, \rho) = Vu\left(\frac{S_T}{V}, \rho\right) - VT \frac{S_T}{V}$$

$$f(T, \rho) = u(s_T, \rho) - Ts_T,$$

where

$$s_T := \frac{S_T}{V}$$

and $S_T = S_T(V, N)$ is the unique solution to

$$T_0(S_T(V, N), V, N) = T.$$

Since T is homogeneous of degree zero,

$$\begin{aligned} T &= T_0(S_T(V, N), V, N) \\ &= T_0(s_T, 1, \rho) \\ &= T_0(s_T, \rho) \end{aligned}$$

Thus, we have proven the following:

Theorem (8.1): The free energy density satisfies

$$(8.4) \quad f(T, \rho) = u(s_T, \rho) - Ts_T,$$

where $s_T = s_T(\rho)$ is the unique solution to

$$(8.5) \quad T_0(s_T(\rho), \rho) = T.$$

Theorem (8.2): We have

$$dF = -SdT - PdV + \sum_{j=1}^r \mu_j dN_j$$

by which we mean $F = F(T, V, \vec{N})$ and

$$(8.6) \quad \frac{\partial F}{\partial T}(T, V, \vec{N}) = -S_T(V, \vec{N}),$$

$$(8.7) \quad \frac{\partial F}{\partial V}(T, V, \vec{N}) = -P_U(S_T(V, \vec{N}), V, \vec{N}),$$

$$(8.8) \quad \frac{\partial F}{\partial N_j}(T, V, \vec{N}) = \mu_{U,j}(S_T(V, \vec{N}), V, \vec{N}),$$

where $S_T = S_T(V, \vec{N})$ is the unique solution of the equation
 $T_U(S_T(V, \vec{N}), V, \vec{N}) = T.$

Proof: Exercise. ///

Example (8.3): Suppose that

$$\tilde{S}(U, V, N) = \left(\frac{NVUR^2}{v_0 \Theta} \right)^{1/3},$$

where $R, v_0, \Theta > 0$ are constants.

Then,

$$\tilde{U}(S, V, N) = \frac{\nu_0 \theta S^3}{N V R^2},$$

as we have observed many times, and

$$\begin{aligned} T_0(S, V, N) &= \frac{\partial \tilde{U}}{\partial S} \\ &= \frac{3 \nu_0 \theta S^2}{N V R^2}. \end{aligned}$$

It follows that, for any $T \geq 0$

$$S_T(V, N) = \left(\frac{N T V R^2}{3 \nu_0 \theta} \right)^{1/2}.$$

Therefore,

$$\begin{aligned} F(T, V, N) &= \tilde{U}(S_T(V, N), V, N) - T S_T(V, N) \\ &= \frac{\nu_0 \theta (S_T(V, N))^3}{N V R^2} - T S_T(V, N) \\ &= \frac{\nu_0 \theta}{N V R^2} \left(\frac{N T V R^2}{3 \nu_0 \theta} \right)^{3/2} \\ &\quad - T \left(\frac{N T V R^2}{3 \nu_0 \theta} \right)^{1/2} \end{aligned}$$

$$= \left(\frac{NTVR^2}{3v_0\theta} \right)^{1/2} \left\{ \frac{\cancel{v_0\theta} \cancel{NTVR^2}}{\cancel{NR^2} \cancel{3v_0\theta}} - T \right\}$$

$$= -\frac{2}{3} \left(\frac{NTVR^2}{3v_0\theta} \right)^{1/2} T^{3/2}$$

The partial with respect to T is

$$\begin{aligned} \frac{\partial F}{\partial T} &= - \left(\frac{NTVR^2}{3v_0\theta} \right)^{1/2} \\ &= -S_T(V, N). \end{aligned}$$

The pressure is given by

$$P_v(S, V, N) = -\frac{\partial \tilde{U}}{\partial V} = \frac{v_0\theta S^3}{NV^2R^2}.$$

Therefore,

$$\begin{aligned} P_v(S_T(V, N), V, N) &= \frac{v_0\theta}{NV^2R^2} \left(\frac{NTVR^2}{3v_0\theta} \right)^{3/2} \\ &= \left(\frac{T}{3} \right)^{3/2} \left(\frac{NR^2}{v_0V\theta} \right)^{1/2}. \end{aligned}$$

The partial of F with respect to V is

$$\frac{\partial F}{\partial V} = -\frac{1}{3} \left(\frac{NR^2}{3Vv_0\theta} \right)^{1/2} T^{3/2}$$

$$= - \left(\frac{T}{3} \right)^{3/2} \left(\frac{NR^2}{v_0 v \theta} \right)^{1/2}$$

$$= - P_v(S_T(v, N), v, N).$$

The reader should confirm that

$$\frac{\partial F}{\partial N_j} = \mu_{v,j}(S_T(v, N), v, N).$$

Theorem (8.4): We have

$$df = -s dT + \sum_{j=1}^r \mu_j dp_j$$

by which we mean $f = f(T, p)$ and

$$(8.9) \quad \frac{\partial f}{\partial T}(T, \vec{p}) = -s_T(\vec{p}),$$

$$(8.10) \quad \frac{\partial f}{\partial p_j}(T, \vec{p}) = \mu_{v,j}(s_T(\vec{p}), \vec{p}),$$

where $s_T(\vec{p})$ is the unique solution to

$$(8.11) \quad T_v(s_T(\vec{p}), \vec{p}) = T.$$

Proof: Exercise. ///

A good example goes a long way.

Example (8.5): We will use our standard model.
Suppose that

$$\tilde{U}(S, V, N) = \frac{v_0 \theta S^3}{N V R^2}.$$

Then

$$\begin{aligned} u(s, p) &= \frac{1}{V} \tilde{U}(S, V, N) \\ &= \tilde{U}(s, 1, p) \end{aligned}$$

$$= \frac{v_0 \theta s^3}{p R^2}$$

and

$$T_u(s, p) = \frac{3 v_0 \theta s^2}{p R^2}.$$

Thus,

$$s_T(p) = \left(\frac{p R^2 T}{3 v_0 \theta} \right)^{1/2},$$

and

$$f(T, p) = u(s_T(p), p) - T s_T(p)$$

$$= \frac{v_0 \theta (s_T(p))^3}{p R^2} - T \left(\frac{p R^2 T}{3 v_0 \theta} \right)^{1/2}$$

$$= \frac{v_0 \theta}{p R^2} \left(\frac{p R^2 T}{3 v_0 \theta} \right)^{3/2} - T \left(\frac{p R^2 T}{3 v_0 \theta} \right)^{1/2}$$

$$= \left(\frac{p R^2 T}{3 v_0 \theta} \right)^{1/2} \left\{ \frac{\cancel{v_0 \theta}}{\cancel{p R^2}} \frac{\cancel{p R^2} T}{3 \cancel{v_0 \theta}} - T \right\}$$

$$= -\frac{2 T^{3/2}}{3} \left(\frac{p R^2}{3 v_0 \theta} \right)^{1/2}$$

The partial with respect to T is

$$\frac{\partial f}{\partial T}(T, p) = - \left(\frac{p T R^2}{3 v_0 \theta} \right)^{1/2}$$

$$= -s_T(p).$$

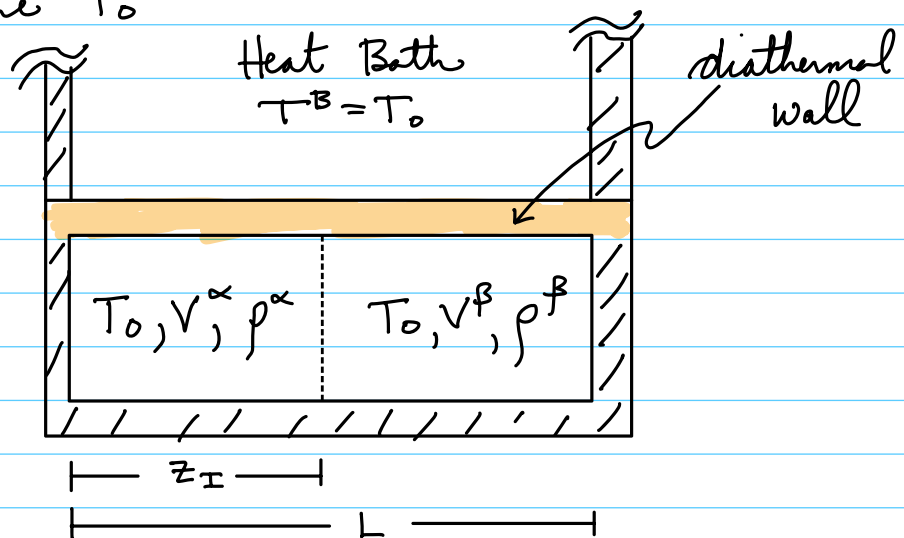
Maxwell's Common Tangent Construction

Here we will review a situation for which the Helmholtz potential is important.

Consider a unary, isothermal, two phase system, for example liquid and solid iron.

We will assume the β phase is the high-density phase (a solid, perhaps) and the α phase is the low-density phase (a liquid, perhaps).

Both phases are in contact with a heat bath at temperature T_0 .



Here the densities, ρ^α and ρ^β , play the roles of the numbers of moles, N^α and N^β .

We will assume, for simplicity, that parameters vary in only one spatial dimension, z .

Suppose the cross-section area of the container is A . Then

$$V^\alpha = A z_I$$

$$V^\beta = A (L - z_I)$$

$$V^\alpha + V^\beta = A \cdot L =: V_0 \quad \left(\begin{array}{l} \text{total volume} \\ \text{constant} \end{array} \right)$$

Now,

$$N^\alpha = V^\alpha \rho^\alpha,$$

$$N^\beta = V^\beta \rho^\beta.$$

Here

$$[\rho^\alpha] = \text{moles / unit vol}$$

$$N_0 = N^\alpha + N^\beta \quad \left(\begin{array}{l} \text{total mole} \\ \text{number const} \end{array} \right)$$

$$= V^\alpha \rho^\alpha + V^\beta \rho^\beta$$

$$= A z_I \rho^\alpha + A (L - z_I) \rho^\beta$$

It will be convenient to introduce

$$\rho_0 = \frac{N_0}{V_0} = \frac{N_0}{L \cdot A}.$$

Hence,

(8.12)

$$\rho_0 \cdot L = z_I \rho^\alpha + (L - z_I) \rho^\beta.$$

Assume that

$$F_{\alpha+\beta}(V^\alpha, V^\beta, p^\alpha, p^\beta) = V^\alpha f^\alpha(p^\alpha) + V^\beta f^\beta(p^\beta).$$

We suppress the temperature dependence, for simplicity. The variables f^α and f^β are called free energy densities. We can eliminate the volume variables using constraints:

$$\begin{aligned} F_{\alpha+\beta}(V^\alpha, V^\beta, p^\alpha, p^\beta) &= V^\alpha f^\alpha(p^\alpha) + V^\beta f^\beta(p^\beta) \\ &= A z_{\pm} f^\alpha(p^\alpha) + A(L - z_{\pm}) f^\beta(p^\beta). \end{aligned}$$

Using the mass constraint (8.12), we have

$$z_{\pm} = \frac{L(p_0 - p^\beta)}{p^\alpha - p^\beta}$$

or

(8.13)

$$z_{\pm} = \frac{L(p^\beta - p_0)}{p^\beta - p^\alpha}$$

Similarly, we have

(8.14)

$$L - z_{\pm} = \frac{L(p_0 - p^\alpha)}{p^\beta - p^\alpha}$$

Thus,

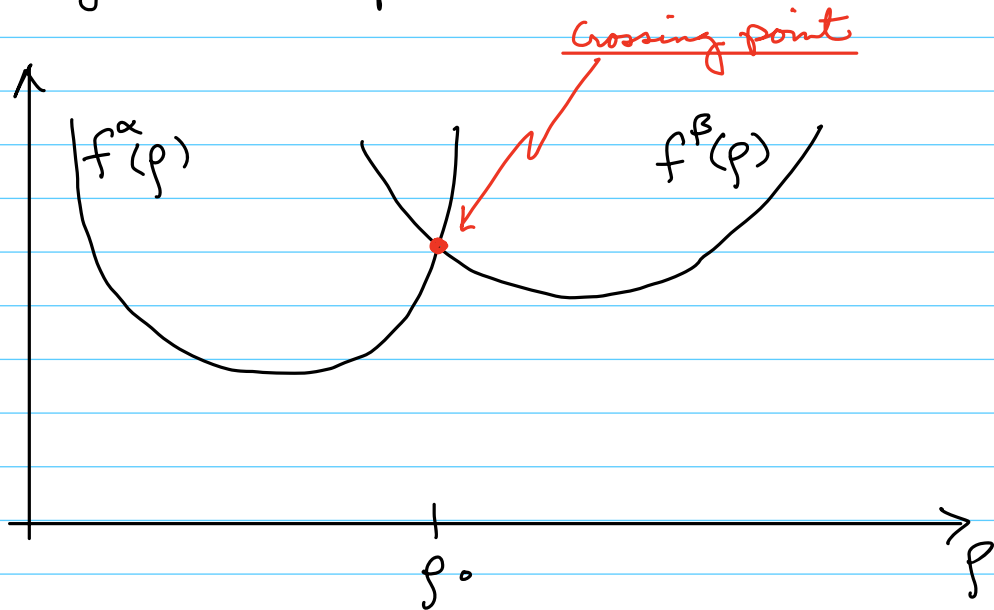
$$F_{\alpha+\beta}(V^\alpha, V^\beta, p^\alpha, p^\beta) = \overbrace{L \cdot A}^{=V_0} \tilde{f}(p^\alpha, p^\beta),$$

where

$$\tilde{f}(p^\alpha, p^\beta) = \frac{p^\beta - p_0}{p^\beta - p^\alpha} f^\alpha(p^\alpha) + \frac{p_0 - p^\alpha}{p^\beta - p^\alpha} f^\beta(p^\beta).$$

Recall, the setting is isothermal, $T^\alpha = T^\beta = T_0$. So equilibrium is characterized by a minimum in the free energy. Thus, we seek a minimizer of $\tilde{f}$.

The setting is as follows:



We want p^α and p^β that minimize $\tilde{F}$.

$$0 = \frac{\partial \tilde{f}}{\partial p^\alpha} (p^\alpha, p^\beta)$$

$$0 = \frac{\partial \tilde{f}}{\partial p^\beta} (p^\alpha, p^\beta)$$

Now,

$$\frac{\partial \tilde{f}}{\partial p^\alpha} = \frac{p^\beta - p_0}{(p^\beta - p^\alpha)^2} f^\alpha(p^\alpha) + \frac{p^\beta - p_0}{p^\beta - p^\alpha} f^{\alpha'}(p^\alpha)$$

$$+ \frac{p_0 - p^\beta}{(p^\beta - p^\alpha)^2} f^\beta(p^\beta)$$

$$= \frac{p^\beta - p_0}{p^\beta - p^\alpha} \left\{ \frac{f^\alpha(p^\alpha) - f^\beta(p^\beta)}{p^\beta - p^\alpha} + f^{\alpha'}(p^\alpha) \right\}$$

and

$$\frac{\partial \tilde{f}}{\partial p^\beta} = \frac{p_0 - p^\alpha}{(p^\beta - p^\alpha)^2} f^\alpha(p^\alpha)$$

$$- \frac{p_0 - p^\alpha}{(p^\beta - p^\alpha)^2} f^\beta(p^\beta) + \frac{p_0 - p^\alpha}{p^\beta - p^\alpha} f^{\beta'}(p^\beta)$$

$$= \frac{p_0 - p^\alpha}{p^\beta - p^\alpha} \left\{ \frac{f^\alpha(p^\alpha) - f^\beta(p^\beta)}{p^\beta - p^\alpha} + f^{\beta'}(p^\beta) \right\}$$

Thus, the minimizer must satisfy

(8.15)

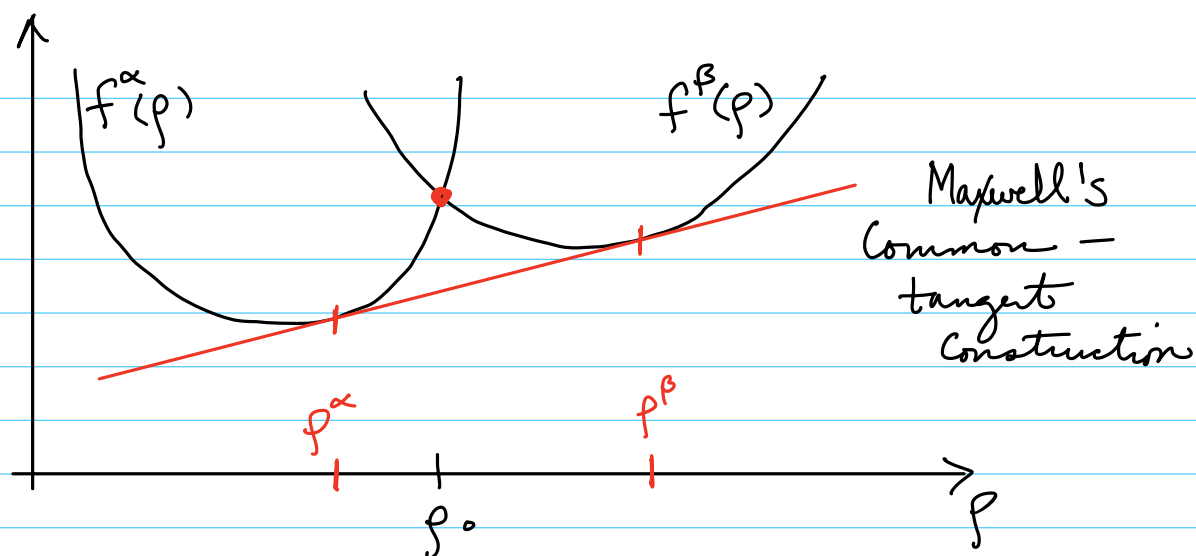
$$\frac{f^\beta(p^\beta) - f^\alpha(p^\alpha)}{p^\beta - p^\alpha} = f^{\alpha'}(p^\alpha)$$

(8.16)

and

$$\frac{f^\beta(p^\beta) - f^\alpha(p^\alpha)}{p^\beta - p^\alpha} = f^{\beta'}(p^\beta)$$

Graphically, these solutions represent the following scenario:



For everything to make sense, we require that

$$\begin{array}{c}
 \text{average density} \\
 \downarrow \\
 \rho^\alpha < \rho_0 < \rho^\beta \\
 \uparrow \qquad \qquad \qquad \uparrow \\
 \text{low density phase} \quad \text{high density phase}
 \end{array}$$

What happens if, say,

$$\rho^\alpha < \rho^\beta < \rho_0,$$

that is, the average is bigger than the density of the high-density phase, β ?

Recall,

$$z_{\pm} = \frac{L(\rho^\beta - \rho_0)}{\rho^\beta - \rho^\alpha}.$$

If $p^\alpha < p^\beta < p_0$, it follows that

$$z_I < 0,$$

which is unphysical.

On the other hand if $p_0 < p^\alpha < p^\beta$, then

$$z_I > L,$$

which is unphysical.

But, if

$$p^\alpha < p_0 < p^\beta,$$

it follows that z_I is fully determined and

$$0 < z_I < L,$$

as desired.

Recall that the chemical potentials satisfy

$$\mu^\alpha = \frac{\partial f^\alpha}{\partial p^\alpha}$$

$$\mu^\beta = \frac{\partial f^\beta}{\partial p^\beta}$$

So that equilibrium occurs when

$$\mu^\alpha = \mu^\beta.$$

Unary, Isothermal Phase Diagrams (at constant volume)

let us consider the following phase diagram

